Chapter 2

Proof of Concept: Polymer-Based Composite Organic Memristive Devices

2.1 Introduction and Motivation

This chapter reports the proof-of-concept polymer-electrolyte memristive platform introduced at the end of chapter 1. The composite combines a semiconducting poly(phenylene vinylene) (PPV) derivative (*Super Yellow*, SY), an oxygen-rich hyperbranched polyester-amide ion-conducting host (*Hybrane*® DEO750 8500), and a lithium salt (lithium triflate, LiOTf, $LiCF_3SO_3$), all processed from a common solvent into a standard two-terminal (2-T) sandwich on indium tin oxide (ITO) and silver. The experiments test whether this common formulation and architecture support current–voltage hysteresis, potentiation and depression, excitatory postsynaptic current, distinct retention regimes, and spike-timing-dependent plasticity. The assays were not all performed on one physical junction: the available specimen mapping is stated in Appendix A. The chapter also

develops a candidate cation–host coordination interpretation. Chapter 4 evaluates its cation-ordering prediction but does not confirm a transferable ordering.

At the start of this work the literature already contained several partial demonstrations of these functions in different architectures (reviewed in section 1.5.2): ion-mediated analogue switching in multi-layer polyaniline junctions [1–3], bistable ionic redistribution in a light-emitting memristor [4], tunable short-to-long-term transitions in a partially-irreversible redox device [5, 6], and reversible analogue control in three-terminal organic electrochemical transistors [7–10]. Each covered part of the specification developed in section 1.2.3 (analogue, reversible, multi-state, electrically driven conductance changes), but their combination in one self-contained two-terminal organic platform with a chemically testable ion-mediated hypothesis remained, at that point, an open problem.

The active-layer logic adopted here is the composite logic of section 1.6.4. The three functional roles (electronic transport, ionic transport, and mobile-charge supply) are assigned to three different components, so that each role can be tuned chemically without redesigning the rest of the stack. *Super Yellow* supplies the electronic-conductance pathway. *Hybrane* supplies an oxygen-rich ion-transport host. LiOTf supplies a small hard cation and a charge-delocalised, weakly coordinating counter-anion. This chemical contrast motivates the candidate mechanism of section 2.5; hard–soft acid–base (HSAB) reasoning alone does not determine mobility in the composite.

The measurements fall into two groups. The first comprises three dynamical protocols: current–voltage hysteresis characteristic of memristive switching, potentiation as a function of the number of applied voltage pulses, and depotentiation as a function of the delay before read-out. These three measurements recur, on the same 2-T device geometry, throughout the comparative work of Chapters 4 and 5. The second

group comprises extended synaptic protocols specific to this chapter: an excitatory post-synaptic current (EPSC) sequence with four numerically separated readouts; the resolution of the depotentiation measurement into short- and longer-lived regimes, denoted short-term memory (STM) and long-term memory (LTM) following the convention of the artificial-synapse literature [11, 12]; and a spike-timing dependent plasticity (STDP) protocol that produces an asymmetric timing kernel. Re-fitting the archived five-junction mean gives branch time constants of approximately 89 ms and 150 ms; these are a waveform-level analogy to biological STDP timescales, not evidence of a biological learning mechanism [13, 14]. The published interpretation associates the fast response with triflate redistribution and the slow response with lithium displacement from oxygen coordination sites. The cation survey of Chapter 4 does not reproduce the resulting simple cation order, so that species-level assignment remains a candidate rather than a conclusion.

Within the thesis the chapter is the only one in which the full synaptic suite is the main evidence; only the three dynamical measurements named above carry forward to the comparative corpus.

2.2 Materials Design and Composite Formulation

2.2.1 Design Rationale

The operating principle of the device is the same one developed in section 1.6.4: ionic migration within a solid-state polymer matrix under an applied electric field redistributes mobile ions near the electrodes, which in turn modifies the energy barriers for charge injection and the doping level of the semiconducting polymer, and therefore the electronic

conductance of the device. The composite active layer is built to satisfy three functional requirements simultaneously:

1. **Electronic-conductance pathway:** a semiconducting polymer provides the medium through which the electronic current flows, and whose conductivity is susceptible to modulation by ionic redistribution and electrochemical doping.

2. **Ion-transport medium:** a polymer electrolyte provides a structural environment in which dissolved ions remain mobile under an applied field, yet are retarded enough by intermolecular interactions that, once the field is removed, the conductance change persists for seconds (long enough to be read out) before fading back toward equilibrium.

3. **Ion source:** a salt provides cations and anions whose field-driven redistribution can modify the electronic current; a species-specific assignment to the two retention regimes is considered as a hypothesis below.

The composition adopted here allows a chemically motivated comparison between the two ions of the salt. Li^+ is a small, hard Lewis acid with a high charge-to-radius ratio ($\lambda = z/r \approx 1.3\,\text{Å}^{-1}$–$1.7\,\text{Å}^{-1}$ across four- to six-fold oxygen coordination, with the corresponding Shannon radii in the range 0.59 Å–0.76 Å [15]). Ether and ester oxygens in *Hybrane* can coordinate alkali cations, while the negative charge of $CF_3SO_3^-$ is delocalised and triflate is commonly treated as a weakly coordinating counter-anion. This contrast motivates a fast-anion/slow-cation hypothesis under bias. It does not establish that assignment: mobility also depends on ion pairing, salt concentration, polymer segmental motion, electrode boundary conditions, and field. The two-regime interpretation below is therefore a testable model, not a direct consequence of HSAB classification.

2.2.2 Super Yellow: The Semiconducting Polymer

Super Yellow (commercial PDY-132, Merck) is the semiconducting PPV-family copolymer in the blend. Its role in the proof-of-concept device is to provide the electronic-conductance path whose injection barriers and local doping state can be modulated by mobile ions. The relevant material facts for the main argument are its solution processability in cyclohexanone, its established use in organic light-emitting devices [16], and frontier levels of approximately $E_{\mathrm{HOMO}} = -5.4\,\mathrm{eV}$ and $E_{\mathrm{LUMO}} = -3.0\,\mathrm{eV}$ [17]. The full copolymer name and component-level notes are collected in Appendix A.

2.2.3 Hybrane DEO750 8500: The Ion-Transport Polymer

Hybrane is an oxygen-rich hyperbranched polyester-amide [18]. In this chapter it is used as the ion-transport host: ether and ester oxygen atoms provide hard Lewis-base coordination sites for alkali cations, while the hyperbranched architecture suppresses crystallisation and keeps the blend amorphous. Ion motion is therefore treated as segmental-motion-assisted hopping between local coordination environments, in the polymer-electrolyte picture introduced in section 1.6.1. The relevant migration activation scale is of order $50\,\mathrm{kJ\,mol^{-1}}$ for polymer electrolytes near room temperature [19]; the longer discussion of host structure and transport is in Appendix A.

2.2.4 Lithium Triflate: The Ionic Salt

LiOTf (lithium trifluoromethanesulfonate, $\mathrm{LiCF_3SO_3}$) supplies the mobile ions. It is compatible with the cyclohexanone-processed *Super Yellow*/*Hybrane* blend and is widely used in polymer-electrolyte and LEC

systems [20, 21]. Its small, hard Li^+ cation can coordinate the host oxygens, whereas the triflate charge is delocalised. The chapter uses that contrast to formulate, and later delimit, a candidate explanation of the two relaxation regimes; supporting material notes are in Appendix A.

2.2.5 Composite Formulation and Preparation

The active layer uses a mass ratio of *Super Yellow*:*Hybrane*:LiOTf = 1:0.30:0.09, processed from cyclohexanone under dry nitrogen. After mixing, the final solution concentrations are $8.72\,\mathrm{mg\,mL^{-1}}$ (*Super Yellow*), $2.62\,\mathrm{mg\,mL^{-1}}$ (*Hybrane*), and $0.78\,\mathrm{mg\,mL^{-1}}$ (LiOTf). The ratio is the optimised formulation reported in the published proof-of-concept work [22]: higher salt contents degraded film homogeneity, whereas lower salt contents reduced the available mobile-carrier density. Detailed solution preparation and optimisation notes are given in Appendix A. The same formulation had also been validated in related *Super Yellow*/*Hybrane*/lithium-salt LECs with long electroluminescent operation [23].

2.3 Device Architecture and Fabrication

2.3.1 Device Geometry: The Two-Terminal Vertical Structure

The device adopts a vertical sandwich geometry, in which the active layer is placed between a bottom transparent electrode and a top metallic electrode (see fig. 2.1). This 2-T architecture is the simplest memristive configuration: it has no gate electrode and needs no transistor to address the device, in contrast to the *three-terminal* (3-T) organic synaptic transistors discussed in section 1.5.2 [7, 8, 24].

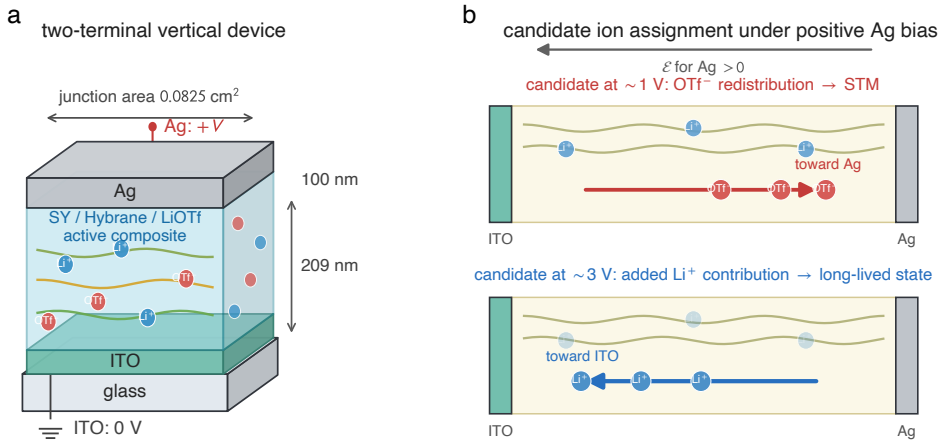


Figure 2.1: Schematic of the two-terminal vertical synaptic device and candidate active-layer mechanism. The device stack is ITO-glass / *Super Yellow*/*Hybrane*/LiOTf composite / Ag; the composite separates the electronic pathway, ion-transport host, and mobile-ion supply. The species labels show the working fast-anion/slow-cation assignment reported in Ref. [22], not a direct measurement of ion motion; the comparative cation data do not establish that assignment as a general law.

The 2-T design reduces fabrication complexity: the device is made by three depositions (ITO substrate, spin-coated active layer, evaporated metal), which makes the same stack compatible with the comparative corpus of Chapter 4. The coexistence of short-term memory and long-term memory without an independent gating signal also supports attribution of the measured memory to the active layer rather than to a gate field acting on a separate channel.

The bottom electrode is a pre-patterned layer of ITO deposited on borosilicate glass. ITO combines optical transparency with adequate electronic conductance ($\sigma \approx 10^4\,\mathrm{S\,cm^{-1}}$) [25], allowing optical inspection of the active layer and characterisation of the device stack by techniques such as UV-Vis spectroscopy and electroluminescence imaging. The work function of ITO ($\Phi_{\mathrm{ITO}} \approx 4.7\,\mathrm{eV}$) is well matched to the ionisation potential of

PPV-family semiconductors [26], facilitating hole injection into the active layer.

The top electrode is silver (Ag), deposited by thermal evaporation. Ag was selected over gold for reasons of cost and practical feasibility; its work function ($\Phi_{\mathrm{Ag}} \approx 4.3\,\mathrm{eV}$) creates a modest electron injection barrier with the *Super Yellow* conduction band, consistent with asymmetric injection conditions often observed in LEC-type structures [27]. The junction active area defined by the shadow mask geometry is $0.0825\,\mathrm{cm}^2$.

2.3.2 Fabrication and Active-Layer Morphology

ITO-patterned glass substrates were cleaned by sequential ultrasonication, dried under nitrogen, and transferred into a nitrogen glovebox. The composite solution was spin-coated at 2000 rpm for 60 s and annealed at 75 °C for 3 h. Contact profilometry gave an active-layer thickness of 209 nm at this spin speed, and tapping-mode AFM showed smooth films with RMS roughness in the range 1 nm–3 nm across the investigated processing window [22]. The spin-speed optimisation, profilometry method, AFM parameters, and supporting morphology evidence are reported in Appendix A.

The top electrode was 100 nm of Ag deposited by thermal evaporation through a shadow mask. The mask defines a junction area of $0.0825\,\mathrm{cm}^2$. Detailed cleaning and evaporation parameters are collected in Appendix A. The complete stack therefore requires solution processing followed by one metal evaporation step.

2.4 Electrical Characterisation

Electrical measurements used a Keithley 2450 SourceMeter in a custom probe station. Unless otherwise stated, measurements were made under nitrogen with voltage applied to the Ag top electrode relative to grounded ITO. Protocol-level details are given in Appendix A.

The characterisation follows the two-family distinction of section 2.1. Sections 2.4.1 to 2.4.3 report the three dynamical measurements carried forward to Chapters 4 and 5; sections 2.4.4 and 2.4.5 then report the device-specific extended protocols, the EPSC multi-state readout and spike-timing dependent plasticity. The short- and long-term memory regimes (short-term memory/long-term memory) are read off the variable-delay depotentiation measurement and so appear with it in section 2.4.3.

2.4.1 Current-Voltage Hysteresis

A defining electrical response of a memristive device is a current–voltage (I–V) hysteresis loop pinched at zero voltage [28–30]. At $V = 0$, the net current is zero regardless of device history, while the *slope* of the I–V curve at zero voltage (i.e., the instantaneous conductance) depends on the history of applied voltage.

Figure 2.2 shows the I–V curves obtained by applying successive triangular voltage sweeps to the device at a sweep rate of $0.25\,\mathrm{V\,s^{-1}}$, with the voltage ramped from 0 V to a maximum of 1.2 V and back to 0 V (single polarity, positive). A waiting period of 10 s was observed between successive sweeps to allow partial ionic relaxation. Ten sequential sweeps are shown.

Progressive conductance trajectory. The current at any fixed voltage *increases* with each successive sweep, in separated increments across this sequence. At 1 V the forward current grows from 0.022 µA on the

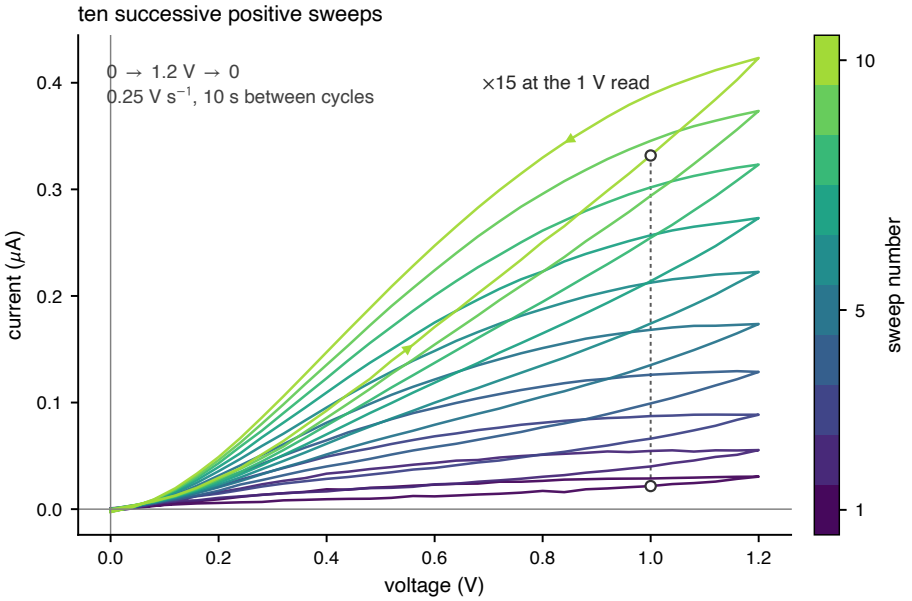


Figure 2.2: Current–voltage hysteresis of the composite memristive device. Ten successive positive triangular sweeps (0 V–1.2 V, 0.25 V s^{-1}, 10 s between cycles) fan out in near-uniform increments, the forward current at 1 V growing roughly 15-fold from the first to the tenth cycle. The clockwise loop closes at the origin (pinched hysteresis) without abrupt switching. Representative SY/Hybrane/LiOTf device (NM_v025) of the published 1:0.30:0.09 formulation, reproducing the hysteresis of Ref. [22].

first sweep to 0.33 µA on the tenth (a ratio of approximately 15 across ten cycles). The increase is +85 % from cycle 1 to cycle 2 and +13 % from cycle 9 to cycle 10. The trace demonstrates graded analogue evolution within one junction and protocol. Without repeated read distributions it does not establish a count of reliably addressable levels. Progressive ionic redistribution is one physical interpretation of the trajectory.

Hysteresis without current rectification. In all cycles, the return sweep (decreasing voltage) produces lower current than the forward sweep (increasing voltage) at the same voltage, generating the characteristic clockwise hysteresis loop for positive voltage sweeps. This is distinct

from simple rectification (diode behaviour): a diode would produce the same forward and reverse I–V curves; the hysteresis here reflects a time-dependent internal state variable (the ionic configuration) that lags the applied voltage.

Absence of abrupt switching. Unlike digital resistance switching (SET/RESET transitions), in which the resistance changes abruptly by orders of magnitude at a threshold voltage, the conductance here evolves continuously and smoothly with each cycle. This *analogue* trajectory is compatible with tuning between the endpoints, but the number, stability, and error rate of independently readable levels were not measured.

Behaviour under negative polarity. Application of negative voltage sweeps (not shown individually in the main figure) produces a qualitatively different hysteresis: the current decreases with successive negative cycles instead of increasing. Electrochemical side reactions near ITO, analogous to insulating-layer formation reported in MEH-PPV-based LECs [21, 31], are one candidate explanation; no chemical or interfacial measurement identifies that process in this device. The quantitative hysteresis comparison therefore uses the positive-polarity protocol.

2.4.2 Variable-Number-of-Pulses Potentiation

Potentiation by a train of write pulses is the second of the three dynamical measurements introduced at the start of this section. Two complementary experiments are reported: a reversible potentiation/depression cycle that establishes the analogue, multi-state character of the conductance, and a measurement of the conductance change as a function of the number of applied pulses.

Reversible Potentiation and Depression

The conductance can be set to a specific value by applying a controlled number of voltage pulses, and decreased back toward the initial value by pulses of opposite polarity, as the potentiation/depression experiment in fig. 2.3a shows.

A pulsed voltage protocol was applied: 50 consecutive write events at 1 V, followed by 50 at −2 V. In the archived waveform each event has an energised dwell of approximately 51 ms followed by a comparable zero-voltage interval, giving a cadence of approximately 0.103 s. The conductance was monitored at the peak of each pulse (i.e., at the 1 V and −2 V levels, respectively). This protocol is analogous to repeated pre-synaptic stimulation and subsequent inhibition.

The conductance is continuously potentiated from its initial value by more than 200 % during the 1 V pulse train, and subsequently depotentiated toward the initial value during the −2 V train. The potentiation and depression curves are smooth and monotonic, without abrupt transitions. Return to a value near the initial level after the depression protocol is consistent with reversible ionic redistribution within the measured protocol.

The write energy is state-dependent because the current rises as the junction potentiates. Estimating each active dwell as $E_k \simeq |V_k I_k| \Delta t_k$ from the archived voltage, current, and time arrays gives 0.084 µJ–1.55 µJ across the 50 positive events (median 1.08 µJ, mean 0.983 µJ, total 49.1 µJ). The first 50 negative events have a median of 0.567 µJ and total 37.3 µJ. The positive-event median corresponds to approximately 131 fJ µm^{-2} over the 0.0825 cm^2 junction. These are junction-level electrical estimates under a zero-order-hold interpretation of the sampled waveform; they exclude source, sensing, and control overhead. They also place this unoptimised large-area proof of concept above the picojoule-to-nanojoule

switching budgets of more mature organic and inorganic devices [32, 33]. The calculation and every event value are regenerated by `scripts/ch2_figures.py`; the summary is in `handouts/ch2_write_energy.csv`.

Quantitative Pulsed Read–Write Protocol

To quantify the potentiation as a function of pulse number, and to separate the act of writing from the act of reading, a non-perturbing read protocol was adopted; the same protocol is reused for the depotentiation measurement of section 2.4.3. The read–write–wait sequence, detailed in Appendix A, comprises three steps:

1. **Read.** A reading pulse ($V_{\mathrm{read}} = 0.5\,\mathrm{V}$) is applied to establish the initial conductance G_0. This reading voltage is intentionally low: the activation threshold for ionic redistribution in this system was determined by independent experiments to lie above 0.7 V [22], so $V_{\mathrm{read}} = 0.5\,\mathrm{V}$ measures the conductance state without perturbing it.

2. **Write.** A train of N_{pulses} write pulses (V_{write}, duration 0.1 s each) is applied.

3. **Wait and read.** After a waiting time t_{wait} under open-circuit (high-impedance) conditions, a second reading pulse is applied to measure the final conductance G_f.

The conductance change is reported as the percentage ratio $\Delta G = (G_f/G_0) \times 100$.

Between experiments, the device is fully reset by connecting the junction to a short circuit for 300 s, which discharges the ionic double layers and restores the homogeneous equilibrium ion distribution. The suitability of this reset protocol was verified by confirming that G_0 returns to its original value within the measurement uncertainty before each new experiment.

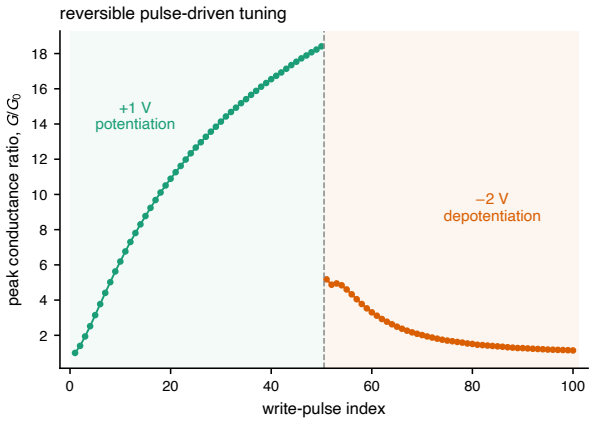
Potentiation as a Function of Pulse Number

Figure 2.3b shows the dependence of ΔG on N_{pulses} for $V_{\text{write}} = 1\,\text{V}$ and $t_{\text{wait}} = 150\,\text{ms}$. The conductance ratio increases monotonically with N_{pulses}, reaching values in excess of $10^3\%$ for $N_{\text{pulses}} > 10^3$. The relationship is approximately linear in log-log space for $N_{\text{pulses}} < 100$ and saturates at large pulse numbers, consistent with a scenario in which the degree of ionic asymmetry that can be built up under a given write voltage is bounded by the total number of mobile ionic carriers.

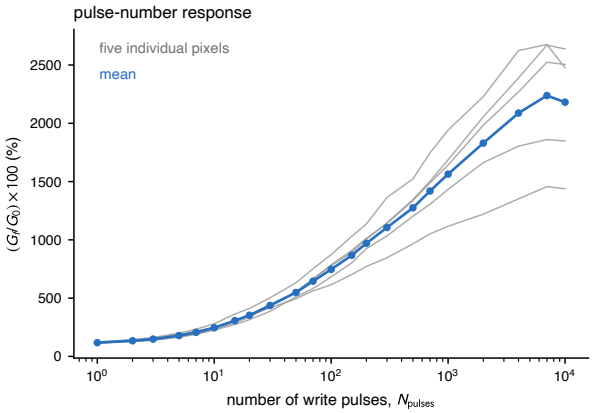
Within this archived pulse-train dataset, the low-pulse-count response varies by approximately 10 % across junctions on one substrate: for the five pixels in fig. 2.3b, the median relative standard deviation is 8.9 % and the largest value is 11.8 % over the points with $N_{\text{pulses}} < 100$. The dispersion increases modestly at large pulse numbers. These values establish within-substrate junction repeatability only; they do not estimate independent-device or batch variability.

2.4.3 Variable-Delay Depotentiation: Short- and Long-Term Retention

The third dynamical measurement carried through the thesis probes how a written conductance state fades once the drive is removed: the depotentiation of the device measured as a function of the delay before read-out. This protocol determines whether the device can emulate short-lived and longer-lived synaptic states on experimentally accessible timescales. In the artificial-synapse literature these regimes are commonly labelled short-term memory and long-term memory; in the present system, the longer-lived regime remains on the sub-minute scale and should therefore be understood operationally, not as a claim of indefinitely persistent storage.



(a) Reversible potentiation/depression.



(b) Pulse-number response.

Figure 2.3: Pulse-driven conductance integration in the composite memristive platform. **(a)** Fifty consecutive 1 V write pulses continuously potentiate junction L5 on NM_v055; subsequent −2 V pulses depotentiate it toward the initial state. **(b)** Dependence of the conductance ratio $(G_f/G_0) \times 100\%$ on the number of write pulses for $V_{\text{write}} = 1\,\text{V}$ and $t_{\text{wait}} = 150\,\text{ms}$. Thin grey lines are five crossbar junctions on the same NM_v055 substrate; the bold line is their mean. The spread therefore measures within-substrate junction variability, not device- or batch-level reproducibility. Data regenerated from the local archive associated with Ref. [22].

The measurement reuses the pulsed read–write protocol of section 2.4.2, with the waiting time t_{wait} now the swept variable and the pulse number held fixed. Figure 2.4 shows the dependence of ΔG on t_{wait} for three excitation conditions: (i) $V_{\mathrm{write}} = 1\,\mathrm{V}$, $N_{\mathrm{pulses}} = 10$ (STM, blue); (ii) $V_{\mathrm{write}} = 1\,\mathrm{V}$, $N_{\mathrm{pulses}} = 50$ (STM, red); and (iii) $V_{\mathrm{write}} = 3\,\mathrm{V}$, $N_{\mathrm{pulses}} = 10$ (LTM, black).

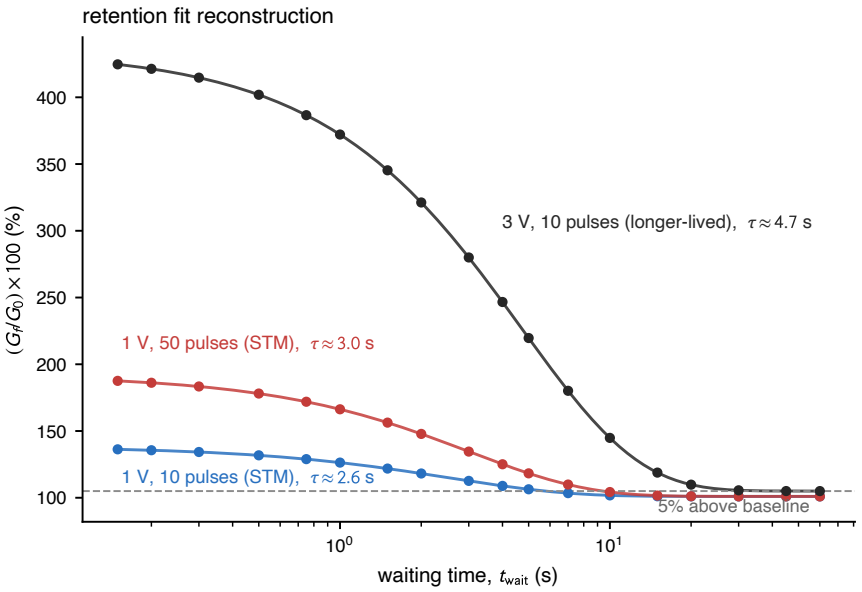


Figure 2.4: Memory retention kinetics. Conductance ratio $(G_f/G_0) \times 100\%$ versus waiting time t_{wait} for STM conditions (1 V; 10 and 50 pulses, blue/red) and the longer-lived regime denoted LTM (3 V, 10 pulses, black). Solid lines are standard-exponential reconstructions of the form in eq. (2.1); the fit parameters are summarised in Appendix A. Data and fit parameters from Ref. [22].

For short waiting times ($t_{\mathrm{wait}} < 500\,\mathrm{ms}$ for STM conditions, $t_{\mathrm{wait}} < 2\,\mathrm{s}$ for LTM conditions), the conductance first *increases* slightly relative to the value immediately after the write pulse train. This counterintuitive behaviour is *not* ascribed to ionic inertia: ion transport in the polymer matrix is strongly overdamped, so any momentum imparted by the field relaxes on a timescale orders of magnitude shorter than the observed

enhancement. It is instead attributed to the continued redistribution of ions after the drive is removed. At the end of the pulse train the ionic configuration and its self-consistent space-charge field have not yet reached their quasi-steady profile; the residual internal field, together with the slow advance of the electrochemically doped region near the electrode, continues to raise the conductance for a brief interval before the diffusion-driven relaxation sets in. An analogous transient enhancement following a stimulus train has been described as "post-tetanic potentiation" in biological synaptic preparations [34, 35].

For longer waiting times, the expected exponential relaxation of conductance is observed as the ionic configuration relaxes back toward equilibrium by thermal diffusion. The fit record summarised in Appendix A reports least-squares fits in IGOR Pro 8 to a standard exponential:

$$\frac{G_f(t_{\mathrm{wait}})}{G_0} = R_\infty + A\exp\!\left(-\frac{t_{\mathrm{wait}}}{\tau}\right) \tag{2.1}$$

where τ is the characteristic relaxation time, A is the decaying amplitude, and R_∞ is the long-waiting-time conductance ratio. Equivalently, the published proof-of-concept fit corresponds to a stretched-exponential model with the stretch exponent fixed at $\beta = 1$. No free β, confidence interval, or goodness-of-fit statistic was reported for these three traces, so they establish only the two experimentally resolved relaxation times, not a fitted distribution of relaxation barriers; the free-exponent Kohlrausch descriptor is reserved for chapter 4, where the DELAYTIME archive is refitted consistently across the corpus.

The fitted characteristic times are $\tau_S = 2.57\,\mathrm{s}$ for the 1 V, 10-pulse STM trace, $\tau_S = 3.01\,\mathrm{s}$ for the 1 V, 50-pulse STM trace, and $\tau_L = 4.73\,\mathrm{s}$ for the 3 V, 10-pulse LTM trace. The corresponding offsets and amplitudes are tabulated in Appendix A. The retention time (defined operationally as the waiting time at which G_f/G_0 returns to within 5 % of the baseline,

$G_f/G_0 < 1.05$) is 10 s–15 s for STM and exceeds 45 s for the longer-lived regime labelled LTM. At event intervals shorter than these windows, the device state can retain information from multiple sequential inputs. The measured retention remains far shorter than that of a conventional non-volatile memory.

The molecular mechanism underlying the STM/LTM distinction is discussed in detail in section 2.5.

2.4.4 Excitatory Post-Synaptic Current

The EPSC measurement is a standard protocol for reporting pulse-conditioned current states in artificial synaptic devices [11, 12]. Here it establishes the response of one reported sequence; the recovered archive contains no repeated state distributions from which to estimate read-error rates, stability, or level capacity. The nomenclature is a functional analogy to a biological experiment, not an assertion of biological equivalence.

The measurement protocol is as follows (fig. 2.5a):

1. **Initialisation.** The device is initialised to its ground state S_0 by applying a fixed read-out bias $V_0 = 1\,\mathrm{V}$ for 1 s. This bias differs in purpose from the sub-threshold probe ($V_{\mathrm{read}} = 0.5\,\mathrm{V}$) used for the non-perturbing read-out of sections 2.4.2 and 2.4.3: the EPSC states are defined operationally by the current response at a single common read-out bias against which all states S_0–S_4 are referenced, not by a passive probe required to leave the state untouched.

2. **Excitation.** A short positive write pulse of 2 V for 50 ms is applied; the device is subsequently held at 1 V for 1 s, during which the conductance spontaneously relaxes to a new excited steady state S_1. The same sequence is repeated to produce a second excited state S_2.

3. **Inhibition.** Two negative write pulses of $-2.5\,\mathrm{V}$ for $50\,\mathrm{ms}$ are applied, producing inhibitory states S_3 and S_4.

The four excited-state readouts are numerically separated from each other and from the ground-state value in the reported sequence. Defining the EPSC readout ratio $R_n = I_{S_n}/I_{S_0}$ as the signed current response of state S_n normalised to that of the ground state S_0 under the fixed read-out bias used in the experiment, the values obtained are: $R_1 = 7.33$, $R_2 = 15.55$, $R_3 = -3.72$, $R_4 = -16.97$. (The ratio is formed from signed currents, not conductances, so the inhibitory states carry a negative sign; conductance itself remains non-negative throughout.) These separations demonstrate the programmed signed response in that sequence. They do not establish reliable state identification under noise or imply additional resolvable levels without repeated writes and reads, retention distributions, and an explicit decision criterion.

The states S_1 and S_2 are reached by positive-polarity stimulation, while S_3 and S_4 are reached by negative-polarity stimulation. Calling these responses excitatory and inhibitory is an operational analogy to long-term potentiation and depression at biological synapses. Both signs are programmed in the same 2-T architecture without an independent gate terminal.

2.4.5 Spike-Timing Dependent Plasticity

Spike-timing dependent plasticity is a timing-dependent Hebbian learning rule: it implements Hebb's local correlation principle [36] in a form that was later observed experimentally in biological preparations [13, 37]. Here the timing delay is defined as $\Delta t = t_{\mathrm{post}} - t_{\mathrm{pre}}$. With this convention, if the presynaptic neuron fires shortly *before* the postsynaptic neuron (causal ordering, $\Delta t < 0$), the synapse is potentiated (LTP); if the presynaptic neuron fires shortly *after* the postsynaptic neuron

(anti-causal ordering, $\Delta t > 0$), the synapse is depressed (LTD). The magnitude of the change decays exponentially with $|\Delta t|$. This temporal asymmetry is essential for the synaptic encoding of causal relationships in neural circuits and underlies the ability of STDP-based learning rules to implement unsupervised feature extraction and pattern recognition [38].

Experimental Protocol

The STDP experiment emulates pre- and postsynaptic spikes by programming their signed resultant across the two-terminal junction. With each template referenced to its own nominal firing time, the applied device voltage is

$$V_{\text{device}}(t, \Delta t) = V_{\text{pre}}(t) + V_{\text{post}}(t - \Delta t) \tag{2.2}$$

Each signed spike template has maximum amplitude $V_{sp} = 1.55\,\text{V}$ and duration 1.2 s. The delay Δt was varied from $-600\,\text{ms}$ to $600\,\text{ms}$. Their overlap makes the resultant depend on timing; the archived applied waveform reaches a transient magnitude of $3.1\,\text{V}$ at $|\Delta t| = 50\,\text{ms}$. Thus $1.55\,\text{V}$ is the template amplitude, not a bound on the voltage experienced by the device.

For $\Delta t < 0$ (pre fires before post), the two spike waveforms partially overlap in time with additive polarity, producing a composite pulse whose peak voltage exceeds V_{sp} and which is directed in the positive sense. The ions then redistribute toward the ITO electrode, the conductance increases, and the device records a net synaptic potentiation.

For $\Delta t > 0$ (post fires before pre), the overlap produces an anti-causal summation. Depending on the exact waveforms, the effective net voltage can be of reduced amplitude or of reversed polarity relative to the causal case, so the ionic redistribution is either weaker or reversed and the

conductance decreases, in the form of a net synaptic depression.

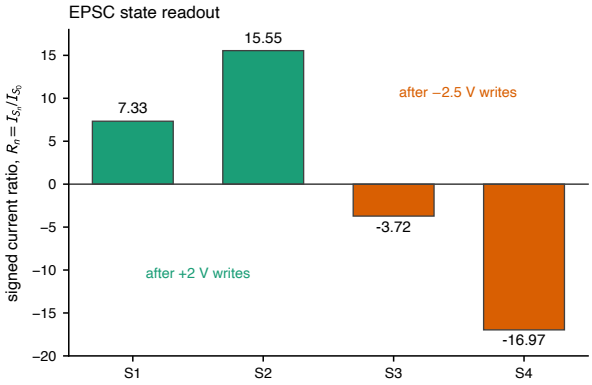
STDP Curve and Fitting

Figure 2.5b shows the measured STDP function: the percentage conductance change ΔG plotted against the timing delay Δt. Under the sign convention above, the measured kernel is asymmetric Hebbian: negative Δt (causal) produces positive ΔG (potentiation), and positive Δt (anti-causal) produces negative ΔG (depression). Each branch (potentiation for $\Delta t < 0$, depression for $\Delta t > 0$) is fitted separately, as a function of the absolute delay $|\Delta t|$, by a decaying exponential:

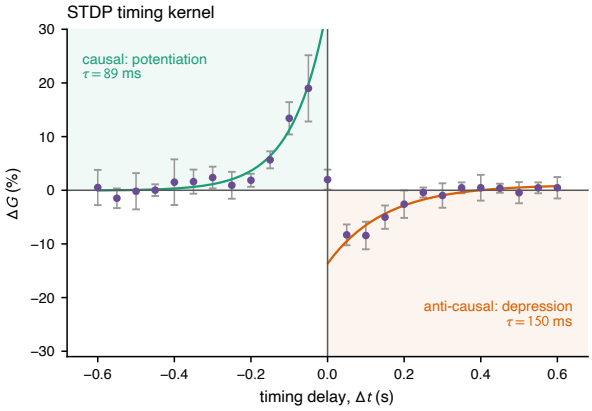
$$\Delta G(\Delta t) = A_{\pm} \exp\left(-\frac{|\Delta t|}{\tau_{\pm}}\right) + G_0 \tag{2.3}$$

where $A_{\pm}$ and $\tau_{\pm}$ are the branch-specific amplitude and time constant ($+$ for the potentiation branch, $-$ for the depression branch) and G_0 is a constant offset. A direct least-squares refit of the archived mean across five junctions gives $\tau_{+} = 89\,\text{ms}$ for potentiation and $\tau_{-} = 150\,\text{ms}$ for depression. Resampling those five same-substrate junctions gives 95% bootstrap intervals of 78 ms–95 ms and 134 ms–172 ms, respectively. These intervals quantify junction spread on one substrate; they are not confidence intervals for an independently fabricated device population. Both branches occupy the tens-to-hundreds-of-milliseconds scale reported in biological STDP experiments [13, 14], but matching a timescale does not make the artificial waveform or ionic mechanism biological.

That the learning rule is *asymmetric Hebbian* reflects the specific voltage waveform design, the stated definition of Δt, and the polarity conventions of the electrode configuration. Different waveform designs or an opposite sign convention would be expected to produce differently labelled STDP kernels, offering an additional degree of tunability in the design of



(a) EPSC state ratios.



(b) STDP timing kernel.

Figure 2.5: Compact summary of the extended synaptic measurements. **(a)** Signed EPSC readout ratios $R_n = I_{S_n}/I_{S_0}$ for the four demonstrated states, reconstructed from the published values; the recovered archive does not resolve their specimen identifier. **(b)** Spike-timing dependent plasticity: mean percentage conductance change ΔG across five junctions on the NM_v055 substrate as a function of the pre/post spike-timing delay Δt, with each branch re-fitted to the decaying exponential of eq. (2.3); error bars are one standard deviation across junctions. The resulting time constants are 89 ms (potentiation) and 150 ms (depression). Data and reported state ratios from Ref. [22]; STDP values and fits regenerated from the local NM_v055 Day19 archive.

hardware neural network architectures built from these devices.

2.5 Theoretical Framework: Ion Transport and the Origin of Memristance

Two limiting-case models from the polymer light-emitting electrochemical cell (LEC) literature provide candidate descriptions of injection and bulk transport: an *electrodynamic* (injection-limited) model [27, 39] and an *electrochemical-doping* (ohmic) model [20, 40, 41], which van Reenen *et al.* [42] reconciled as regimes of field-driven ion redistribution. Their assumptions are set out in Appendix A; neither was discriminated experimentally in the present junctions.

For the present SY/Hybrane/LiOTf system, both limiting cases could contribute. The published working interpretation maps the rapid response to anion redistribution and an injection-limited contribution, and the larger, slower response to Li^+ redistribution and bulk electrochemical doping. The electrical measurements in section 2.4 identify neither the moving species nor the relative ED/ECD contributions. Such discrimination would require, for example, ion-selective or spectroelectrochemical measurements and matched temperature-, thickness-, or sweep-rate dependence. Moreover, the later cation series shifts the dynamics but runs opposite to the simple coordination-strength order under triflate and reorders with the anion (chapter 4). The mapping is therefore retained as a candidate interpretation, not an established mechanism.

2.5.1 Candidate Coordination Rationale and Its Limits

The candidate explanation for the two memory regimes starts from the coordination reasoning developed in sections 1.6.1 and 1.6.4, applied to the specific salt and host of this device.

Within the HSAB framework, Li^+ is a small hard Lewis acid and ether oxygen is a hard Lewis base [43]. In crystalline polyether complexes, Li^+ can be coordinated by four to six oxygen donors (five in PEO_3:$LiCF_3SO_3$: three ether oxygens and two triflate oxygens), with oxygen–lithium distances of 1.9 Å–2.1 Å [44]. Those structural facts support the plausibility of cation–oxygen coordination. They do not provide the migration barrier in the amorphous ternary *Super Yellow*/*Hybrane*/LiOTf film, and a local binding preference must not be equated with a macroscopic drift mobility.

Triflate is charge-delocalised and commonly described as weakly coordinating in polymer electrolytes [45, 46]. That makes greater anion mobility plausible, but does not determine it in this blend: ion pairing, polymer segmental motion, concentration, morphology, electrode reactions, and field can change both transference and activation barriers.

The resulting working hypothesis assigns the two drive regimes as follows:

- A voltage of 1 V across a 209 nm film corresponds to approximately $4.8\,\mathrm{MV\,m^{-1}}$. The candidate assignment associates the resulting 10 s–15 s retention with predominantly triflate redistribution; ion motion was not observed directly.

- A voltage of 3 V corresponds to approximately $14.4\,\mathrm{MV\,m^{-1}}$. The candidate assignment associates the retention beyond 45 s with an added cation contribution, consistent with stronger Li^+–oxygen coordination but not uniquely implied by it [44, 45].

This account yields a specific prediction: at fixed host, anion, composition, electrode, and protocol, the slow fading-memory timescale should shorten along $Li^+ \rightarrow Na^+ \rightarrow K^+$. Chapter 4 evaluates that prediction in a PEO-based survey. The measured triflate retention runs in the opposite order, and the ordering changes with the anion; the relevant chemistry cells are also sample-limited. The comparative evidence therefore does not corroborate a general cation–coordination law. It leaves the species assignment in this proof-of-concept device unresolved and identifies the matched, replicated experiment required to test it.

2.6 Device Stability and Reproducibility

2.6.1 Ambient Shelf Stability

Two distinct notions of device longevity must be kept separate. *Cycling endurance* (the number of write/erase cycles a device sustains before its performance parameters drift outside specified tolerances) governs suitability for the millions-to-billions of weight updates of on-chip training; it is not characterised for the present devices and is left for future work. *Shelf (calendar) stability* (the retention of performance during storage and intermittent operation under ambient conditions) is what is reported here, the more immediately relevant metric at the proof-of-concept stage, where the question is whether the unencapsulated composite survives realistic handling at all.

The devices reported here were characterised at regular intervals over a period of two weeks ($\sim$300 h) after fabrication, without encapsulation, while stored in ambient air at room temperature. The monitored metrics were the baseline conductance G_0, the maximum achievable conductance change $\Delta G_{\max}$, and the STM retention time τ_S; the supporting stability record is collected in Appendix A.

All three metrics remain within 20 % of their initial values over the full two-week measurement period [22]. This stability compares favourably with other organic memristive devices reported in the literature, which typically exhibit significant degradation within hours to days under ambient conditions [47]. The enhanced stability is attributed to the solid-state nature of the composite: unlike devices based on liquid electrolytes or gel polymers, the Hybrane matrix provides a mechanically cohesive environment that protects the ions from evaporation and the polymer films from delamination, consistent with the stability of related Hybrane-based LEC blends [23].

2.6.2 Within-Substrate Junction Repeatability

Reproducibility is a persistent challenge for memristive devices, particularly those based on filamentary switching in inorganic materials, where the stochastic nucleation of conducting filaments produces large device-to-device and cycle-to-cycle variability [48]. Three evidence units must be distinguished here: repeated cycles on one junction, different junctions on one patterned substrate, and independently fabricated devices or batches. The archived proof-of-concept data quantify only the first two.

The strongest repeatability statement available from the archived proof-of-concept data is the one attached to the pulse-number experiment in fig. 2.3b. That dataset contains five crossbar junctions on the NM_v055 substrate, measured under the same 1 V write and 150 ms wait protocol. The shaded band is one standard deviation across those five junctions. For $N_{\mathrm{pulses}} < 100$, the median RSD is 8.9 % and the largest RSD is 11.8 %; the appropriate claim is therefore within-substrate junction repeatability at about the 10 % level in the low-pulse-count regime, not an independent-device reproducibility estimate.

The EPSC readout and STDP measurements are retained as protocol

demonstrations, but the recovered archive does not contain a normalized EPSC population table comparable to the PULSES and DELAYTIME tables used later in the thesis. They are therefore not used to support an independent EPSC RSD claim here. Likewise, the published proof-of-concept data establish same-substrate junction repeatability and two-week shelf stability; they do not constitute a powered device-to-device or batch-to-batch reproducibility study. Campaign-scale reproducibility (a separate metric on a separate axis) and its resolution by a change of ion-transport host are taken up in chapter 3.

2.7 Comparison at the Time of Publication

The historical comparison table reproduced in Appendix A places this platform alongside representative organic and hybrid memristive devices available at the time of the 2022 publication. None of the devices in that set combines fully reversible operation with both short-term memory- and long-term memory-labelled regimes in the same 2-T formulation and architecture. The closest prior report, Liu *et al.* [5], achieved short-to-long-term transitions through irreversible redox chemistry [6], whereas here the conductance returns to G_0 after the reset protocol. The table also contains no other example of STDP in a 2-T architecture, where the timing rule emerges from the superposition of pre- and postsynaptic pulses at the Ag terminal rather than from the gate of a 3-T device [7, 11, 49]. On the operating-regime axes the platform is competitive in unencapsulated shelf lifetime ($\sim$300 h, roughly an order of magnitude beyond comparable organic devices [47]) but not in energy per event: the archived positive train has a median of 1.08 µJ, above CMOS and mature inorganic-memristor budgets [33, 50]. The full table and a per-axis discussion are given in Appendix A.

The comparison above retains the device-level metrics conventional in the

organic memristive literature of the early 2020s: reversibility, retention window, energy per event, and operational lifetime. Subsequent work has expanded the field through organic electrochemical neurons and neuromorphic circuits [51, 52] and the wider organic-iontronic memristor field [53], so the table locates the 2022 result historically. Within the thesis argument those metrics play a narrower role: they establish that the platform is reversible, dynamically addressable, and stable enough to support the comparative study of Chapter 4 (a PEO-based triflate composition and cation sweep, characterised through the same three common dynamical measurements) and the temporal-computing reinterpretation of Chapter 5. The extended synaptic protocols reported here remain specific to this SY/Hybrane/LiOTf device.

2.8 Summary

The fully reversible 2-T organic memristive platform supports two sets of synaptic measurements with the same polymer-composite formulation and stack. I–V hysteresis, variable-number-of-pulses potentiation, and variable-delay depotentiation carry through the rest of the thesis. EPSC, the separated short-term memory/long-term memory reading of the retention data, and STDP remain specific to this proof-of-concept formulation.

A ternary *Super Yellow*/*Hybrane*/LiOTf composite satisfies the three functional requirements of section 2.2.1 (electronic pathway, ion-transport medium, ion source), fabricated as a 2-T ITO/composite/Ag sandwich by solution processing and one evaporation step, with a 209 nm active layer below 3 nm RMS roughness. On this common platform the three dynamical measurements carried forward to the comparative corpus were established: analogue I–V hysteresis (forward current at 1 V growing ∼15-fold over ten cycles); fully reversible pulse potentiation/depression exceeding

200 %, growing past 10^3% for $N_{\text{pulses}} > 10^3$; and variable-delay depotentiation resolving two regimes (short-term memory, $\tau_S = 2.57\,\text{s}$–$3.01\,\text{s}$, retention 10 s–15 s; long-term memory, $\tau_L = 4.73\,\text{s}$, retention $> 45\,\text{s}$, from $\beta = 1$ fits). The archived positive-pulse train consumes 0.084 µJ–1.55 µJ per event as the state changes (median 1.08 µJ). The extended protocols add one numerically separated four-state EPSC sequence (signed ratios −17 to +16, without a population-level read-error estimate) and an asymmetric-Hebbian STDP window whose re-fitted branch constants are 89 ms and 150 ms. A fast-triflate/slow-lithium account and the ED/ECD limiting cases provide a candidate interpretation of the two retention regimes [42] (Appendix A); the electrical data do not identify the mobile species, and the later cation survey does not confirm the predicted simple ordering. The published shelf record spans ∼300 h unencapsulated in ambient conditions (cycling endurance not characterised), while the NM_v055 pulse data show ∼10 % low-pulse-count variability across five same-substrate junctions (median RSD 8.9 %); campaign-scale reproducibility is taken up in chapter 3.

The SY/Hybrane/LiOTf formulation and 2-T architecture provide the proof-of-concept platform for the replicated test of cation–polyether coordination in chapter 4. Composition is the replicated quantitative axis in that PEO-based corpus, while host, anion, and cation substitutions remain sample-limited; the data do not establish a host- and anion-independent Li/Na/K law. The EPSC, separated short-term memory/long-term memory retention, and STDP measurements reported here serve only as formulation-level lithium priors in the later chapters. Chapter 5 uses behavioural models fitted to the comparative corpus to test the computational value of the measured fading memory.

References

1. Erokhin, V., Berzina, T. & Fontana, M. P. Hybrid electronic device based on polyaniline-polyethyleneoxide junction. *Journal of Applied Physics* **97,** 064501. doi:10.1063/1.1861508 (2005).

2. Smerieri, A., Berzina, T., Erokhin, V. & Fontana, M. P. Polymeric electrochemical element for adaptive networks: Pulse mode. *Journal of Applied Physics* **104,** 114513. doi:10.1063/1.3033399 (2008).

3. Berzina, T., Smerieri, A., Bernabò, M., *et al.* Optimization of an organic memristor as an adaptive memory element. *Journal of Applied Physics* **105,** 124515. doi:10.1063/1.3153944 (2009).

4. Zakhidov, A. A., Jung, B., Slinker, J. D., Abruña, H. D. & Malliaras, G. G. A light-emitting memristor. *Organic Electronics* **11,** 150–153. doi:10.1016/j.orgel.2009.09.015 (2010).

5. Liu, G., Wang, C., Zhang, W., *et al.* Organic Biomimicking Memristor for Information Storage and Processing Applications. *Advanced Electronic Materials* **2,** 1500298. doi:10.1002/aelm.201500298 (2016).

6. Heremans, P., Gelinck, G. H., Müller, R., Baeg, K.-J., Kim, D.-Y. & Noh, Y.-Y. Polymer and Organic Nonvolatile Memory Devices. *Chemistry of Materials* **23,** 341–358. doi:10.1021/cm102006v (2011).

7. Gkoupidenis, P., Schaefer, N., Garlan, B. & Malliaras, G. G. Neuromorphic Functions in PEDOT:PSS Organic Electrochemical Transistors. *Advanced Materials* **27,** 7176–7180. doi:10.1002/adma.201503674 (2015).

8. Xu, W., Min, S.-Y., Hwang, H. & Lee, T.-W. Organic core-sheath nanowire artificial synapses with femtojoule energy consumption. *Science Advances* **2,** e1501326. doi:10.1126/sciadv.1501326 (2016).

9. Van de Burgt, Y., Lubberman, E., Fuller, E. J., *et al.* A non-volatile organic electrochemical device as a low-voltage artificial synapse for neuromorphic computing. *Nature Materials* **16,** 414–418. doi:10.1038/nmat4856 (2017).

10. Fuller, E. J., Keene, S. T., Melianas, A., *et al.* Parallel programming of an ionic floating-gate memory array for scalable neuromorphic computing. *Science* **364,** 570–574. doi:10.1126/science.aaw5581 (2019).

11. Kim, Y., Chortos, A., Xu, W., *et al.* A bioinspired flexible organic artificial afferent nerve. *Science* **360,** 998–1003. doi:10.1126/science.aao0098 (2018).

12. Van de Burgt, Y., Melianas, A., Keene, S. T., Malliaras, G. & Salleo, A. Organic electronics for neuromorphic computing. *Nature Electronics* **1,** 386–397. doi:10.1038/s41928-018-0103-3 (2018).

13. Bi, G.-q. & Poo, M.-m. Synaptic Modifications in Cultured Hippocampal Neurons: Dependence on Spike Timing, Synaptic Strength, and Postsynaptic Cell Type. *Journal of Neuroscience* **18,** 10464–10472. doi:10.1523/JNEUROSCI.18-24-10464.1998 (1998).

14. Kuzum, D., Yu, S. & Wong, H.-S. P. Synaptic electronics: materials, devices and applications. *Nanotechnology* **24,** 382001. doi:10.1088/0957-4484/24/38/382001 (2013).

15. Shannon, R. D. Revised effective ionic radii and systematic studies of interatomic distances in halides and chalcogenides. *Acta Crystallographica Section A* **32,** 751–767. doi:10.1107/S0567739476001551 (1976).

16. Friend, R. H., Gymer, R. W., Holmes, A. B., *et al.* Electroluminescence in conjugated polymers. *Nature* **397,** 121–128. doi:10.1038/16393 (1999).

17. Zhang, M., Höfle, S., Czolk, J., Mertens, A. & Colsmann, A. All-solution processed transparent organic light emitting diodes. *Nanoscale* **7,** 20009–20014. doi:10.1039/C5NR05820A (2015).

18. Froehling, P. Development of DSM's Hybrane hyperbranched polyesteramides. *Journal of Polymer Science Part A: Polymer Chemistry* **42,** 3110–3115. doi:10.1002/pola.20113 (2004).

19. Gray, F. M. *Polymer Electrolytes* (Royal Society of Chemistry, Cambridge, 1997).

20. Pei, Q., Yu, G., Zhang, C., Yang, Y. & Heeger, A. J. Polymer Light-Emitting Electrochemical Cells. *Science* **269,** 1086–1088. doi:10.1126/science.269.5227.1086 (1995).

21. Wågberg, T., Hania, P. R., Robinson, N. D., Shin, J.-H., Matyba, P. & Edman, L. On the Limited Operational Lifetime of Light-Emitting Electrochemical Cells. *Advanced Materials* **20,** 1744–1749. doi:10.1002/adma.200702595 (2008).

22. Prado-Socorro, C. D., Giménez-Santamarina, S., Mardegan, L., *et al.* Polymer-Based Composites for Engineering Organic Memristive Devices. *Advanced Electronic Materials* **8,** 2101192. doi:10.1002/aelm.202101192 (2022).

23. Mardegan, L., Dreessen, C., Sessolo, M., Tordera, D. & Bolink, H. J. Stable Light-Emitting Electrochemical Cells Using Hyperbranched Polymer Electrolyte. *Advanced Functional Materials* **31,** 2104249. doi:10.1002/adfm.202104249 (2021).

24. Gkoupidenis, P., Koutsouras, D. A. & Malliaras, G. G. Neuromorphic device architectures with global connectivity through electrolyte gating. *Nature Communications* **8,** 15448. doi:10.1038/ncomms15448 (2017).

25. Kim, H., Gilmore, C. M., Piqué, A., *et al.* Electrical, optical, and structural properties of indium–tin–oxide thin films for organic light-emitting devices. *Journal of Applied Physics* **86,** 6451–6461. doi:10.1063/1.371708 (1999).

26. Ramsdale, C. M., Barker, J. A., Arias, A. C., MacKenzie, J. D., Friend, R. H. & Greenham, N. C. The origin of the open-circuit voltage in polyfluorene-based photovoltaic devices. *Journal of Applied Physics* **92,** 4266–4270. doi:10.1063/1.1506385 (2002).

27. De Mello, J. C., Tessler, N., Graham, S. C. & Friend, R. H. Ionic space-charge effects in polymer light-emitting diodes. *Physical Review B* **57,** 12951–12963. doi:10.1103/PhysRevB.57.12951 (1998).

28. Chua, L. O. Memristor – The Missing Circuit Element. *IEEE Transactions on Circuit Theory* **18,** 507–519. doi:10.1109/TCT.1971.1083337 (1971).

29. Adhikari, S. P., Sah, M. P., Kim, H. & Chua, L. O. Three Fingerprints of Memristor. *IEEE Transactions on Circuits and Systems I: Regular Papers* **60,** 3008–3021. doi:10.1109/TCSI.2013.2256171 (2013).

30. Chua, L. O. If it's pinched it's a memristor. *Semiconductor Science and Technology* **29,** 104001. doi:10.1088/0268-1242/29/10/104001 (2014).

31. Fang, J., Matyba, P., Robinson, N. D. & Edman, L. Identifying and alleviating electrochemical side-reactions in light-emitting electrochemical cells. *Journal of the American Chemical Society* **130,** 4562–4568. doi:10.1021/ja7113294 (2008).

32. Xiao, Z. & Huang, J. Energy-Efficient Hybrid Perovskite Memristors and Synaptic Devices. *Advanced Electronic Materials* **2,** 1600100. doi:10.1002/aelm.201600100 (2016).

33. Yang, J. J., Strukov, D. B. & Stewart, D. R. Memristive devices for computing. *Nature Nanotechnology* **8,** 13–24. doi:10.1038/nnano.2012.240 (2013).

34. Zucker, R. S. Short-term synaptic plasticity. *Annual Review of Neuroscience* **12,** 13–31. doi:10.1146/annurev.ne.12.030189.000305 (1989).

35. Zucker, R. S. & Regehr, W. G. Short-term synaptic plasticity. *Annual Review of Physiology* **64,** 355–405. doi:10.1146/annurev.physiol.64.092501.114547 (2002).

36. Hebb, D. O. *The Organization of Behavior: A Neuropsychological Theory* (Wiley, New York, 1949).

37. Markram, H., Lübke, J., Frotscher, M. & Sakmann, B. Regulation of synaptic efficacy by coincidence of postsynaptic APs and EPSPs. *Science* **275,** 213–215. doi:10.1126/science.275.5297.213 (1997).

38. Bichler, O., Querlioz, D., Thorpe, S. J., Bourgoin, J.-P. & Gamrat, C. Extraction of temporally correlated features from dynamic vision sensors with spike-timing-dependent plasticity.

Neural Networks **32,** 339–348. doi:10.1016/j.neunet.2012.02.022 (2012).

39. De Mello, J. C. Interfacial feedback dynamics in polymer light-emitting electrochemical cells. *Physical Review B* **66,** 235210. doi:10.1103/PhysRevB.66.235210 (2002).

40. Smith, D. L. Steady state model for polymer light-emitting electrochemical cells. *Journal of Applied Physics* **81,** 2869–2880. doi:10.1063/1.363966 (1997).

41. Matyba, P., Maturova, K., Kemerink, M., Robinson, N. D. & Edman, L. The dynamic organic p-n junction. *Nature Materials* **8,** 672–676. doi:10.1038/nmat2478 (2009).

42. Van Reenen, S., Matyba, P., Dzwilewski, A., Janssen, R. A. J., Edman, L. & Kemerink, M. A Unifying Model for the Operation of Light-Emitting Electrochemical Cells. *Journal of the American Chemical Society* **132,** 13776–13781. doi:10.1021/ja1045555 (2010).

43. Pearson, R. G. Hard and Soft Acids and Bases. *Journal of the American Chemical Society* **85,** 3533–3539. doi:10.1021/ja00905a001 (1963).

44. Lightfoot, P., Mehta, M. A. & Bruce, P. G. Crystal Structure of the Polymer Electrolyte Poly(ethylene oxide)3:LiCF3SO3. *Science* **262,** 883–885. doi:10.1126/science.262.5135.883 (1993).

45. Bruce, P. G. & Vincent, C. A. Polymer electrolytes. *Journal of the Chemical Society, Faraday Transactions* **89,** 3187–3203. doi:10.1039/ft9938903187 (1993).

46. Frech, R., Chintapalli, S., Bruce, P. G. & Vincent, C. A. Crystalline and Amorphous Phases in the Poly(ethylene oxide)–LiCF3SO3 System. *Macromolecules* **32,** 808–813. doi:10.1021/ma9812682 (1999).

47. Shih, C.-C., Lee, W.-Y. & Chen, W.-C. Nanostructured materials for non-volatile organic transistor memory applications. *Materials Horizons* **3,** 294–308. doi:10.1039/C6MH00049E (2016).

48. Sun, W., Gao, B., Chi, M., *et al.* Understanding memristive switching via in situ characterization and device modeling. *Nature Communications* **10,** 3453. doi:10.1038/s41467-019-11411-6 (2019).

49. Seo, D.-G., Lee, Y., Go, G.-T., *et al.* Versatile neuromorphic electronics by modulating synaptic decay of single organic synaptic transistor: From artificial neural networks to neuro-prosthetics. *Nano Energy* **65,** 104035. doi:10.1016/j.nanoen.2019.104035 (2019).

50. Indiveri, G. & Liu, S.-C. Memory and Information Processing in Neuromorphic Systems. *Proceedings of the IEEE* **103,** 1379–1397. doi:10.1109/JPROC.2015.2444094 (2015).

51. Harikesh, P. C., Tu, D. & Fabiano, S. Organic electrochemical neurons for neuromorphic perception. *Nature Electronics* **7,** 525–536. doi:10.1038/s41928-024-01200-5 (2024).

52. Krauhausen, I., Griggs, S., McCulloch, I., den Toonder, J. M. J., Gkoupidenis, P., van de Burgt, Y., *et al.* Bio-inspired multimodal learning with organic neuromorphic electronics for behavioral conditioning in robotics. *Nature Communications* **15,** 4765. doi:10.1038/s41467-024-48881-2 (2024).

53. Xia, Y., Zhang, C., Xu, Z., *et al.* Organic iontronic memristors for artificial synapses and bionic neuromorphic computing. *Nanoscale* **16,** 1471–1489. doi:10.1039/D3NR06057H (2024).